ICT305 Topics in IT

Blockchain and Digital Currencies

CIHE Australia

Agenda

1. Motivation for blockchain technology
2. Blockchain technology
3. Blockchain digital transformation
4. Impact of blockchain on the world
5. Digital Currencies
6. Future of Digital Currencies

Motivation for blockchain technology

- The need for decentralization is the key motivation behind the blockchain technology.
  - **Decentralization** is achieved by distributing the computation tasks to all the nodes of the blockchain network.
  - **Decentralization** solves several problems of traditional systems; the single point of failure is one such problem.
- The goal of blockchain is to allow digital information to be recorded and distributed but not edited.
- In this way, a blockchain is the foundation for immutable ledgers, or records of transactions that cannot be altered, deleted, or destroyed.

Motivation for blockchain technology

Example: If you had $100 and spent $20,

A traditional database might just update your balance to $80.

A blockchain would keep a record showing:

- Block 1: "Alice received $100"
- Block 2: "Alice sent $20 to Bob"
- Your current balance ($80) is calculated by adding up all transactions

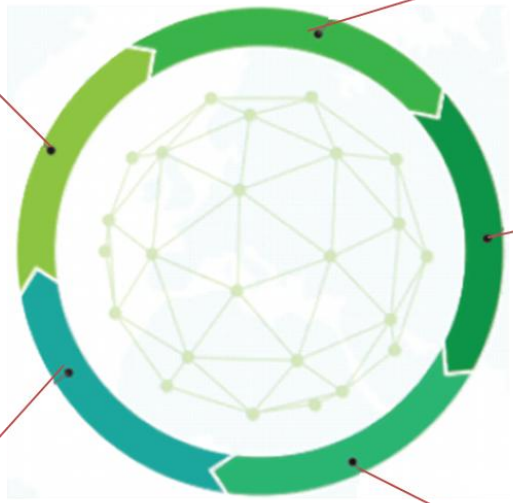
What is blockchain technology

- **The blockchain** is defined as **an open ledger** that offers decentralization to the parties.
  - It also offers transparency, immutability, and security.
  - It has **many features** including being open, distributed, ledger, P2P, and permanent.
- **Blockchain** term was first introduced in the white paper of Bitcoin in 2009 by Satoshi Nakamoto.
- From there, it has come a long way as more and more organizations are interested in it.
- Right now, **Bitcoin** is on its way to implementing a lightning network and other advanced features.

Blockchain

- A Blockchain (BC)- a distributed secure log file or shared ledger with technology to trust transactions without a central authority

A shared ledger technology allowing any participant in the business network to see ledger via distributed consensus.



Every viable transaction is stored in the shared public ledger.

Transactions are placed in blocks, which are linked by one-way hashes.

Each peer address is anonymous and multiple address may map to the same transactor.

Operates in a peer-to-peer mode and is mostly based on DNS and “seed nodes”.

Blockchain Terms

- **Blockchain**:
  - Blockchain is a public digital ledger of past transactions in order.
  - Can be considered as a ledger of bitcoin transactions.
  - The **blockchain** is called so because it is **a chain of blocks**.
  - A blockchain is **a hash-linked data structure**.
- **Block**:
  - **Transactions** are combined into single blocks and every 10 minutes, a new block of about 1MB is size is created.
  - Every block in a blockchain contains 4 components:
    - a timestamp,
    - a reference to the previous block,
    - a summary of the included transaction,
    - the Proof of Work.

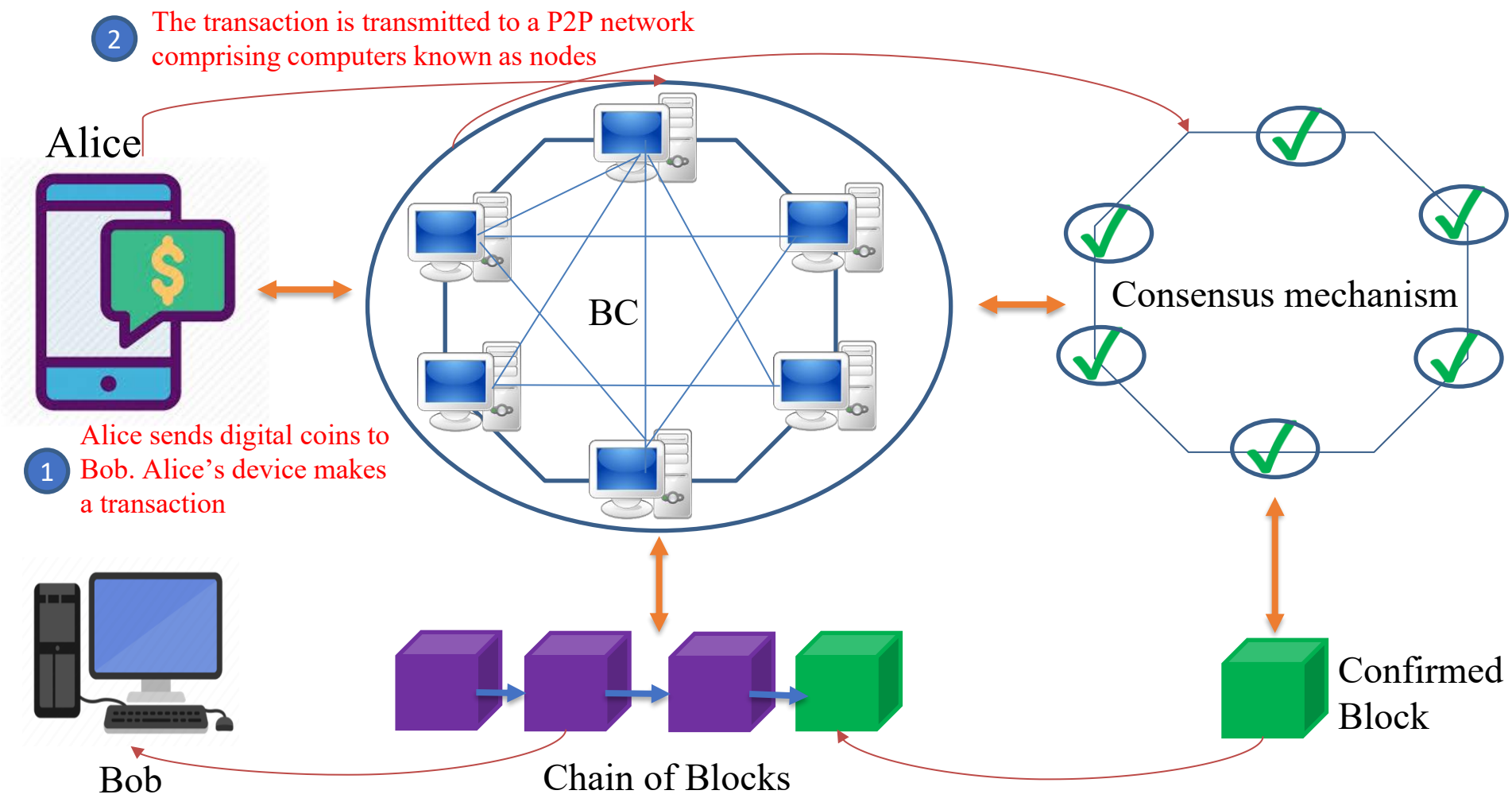
Blockchain terms

- **Mining**:
  - adding transaction records to the blockchain ledger after confirming the validity of the transactions.
  - It involves using complex hardware to perform mathematical calculations in order to verify transactions.
- **Double Spending:**
  - a bitcoin transaction can be sent to two different recipients at the same time is called double spending.
  - This must be avoided at all costs.
- **A Proof of Work:**
  - expensive computations be performed in order to facilitate transactions.
- **Nodes** are distributed computers in the network that all have a copy of the entire blockchain.
- **Bitcoin address**:
  - a long string of alphanumeric characters used to send, receive or hold currency.

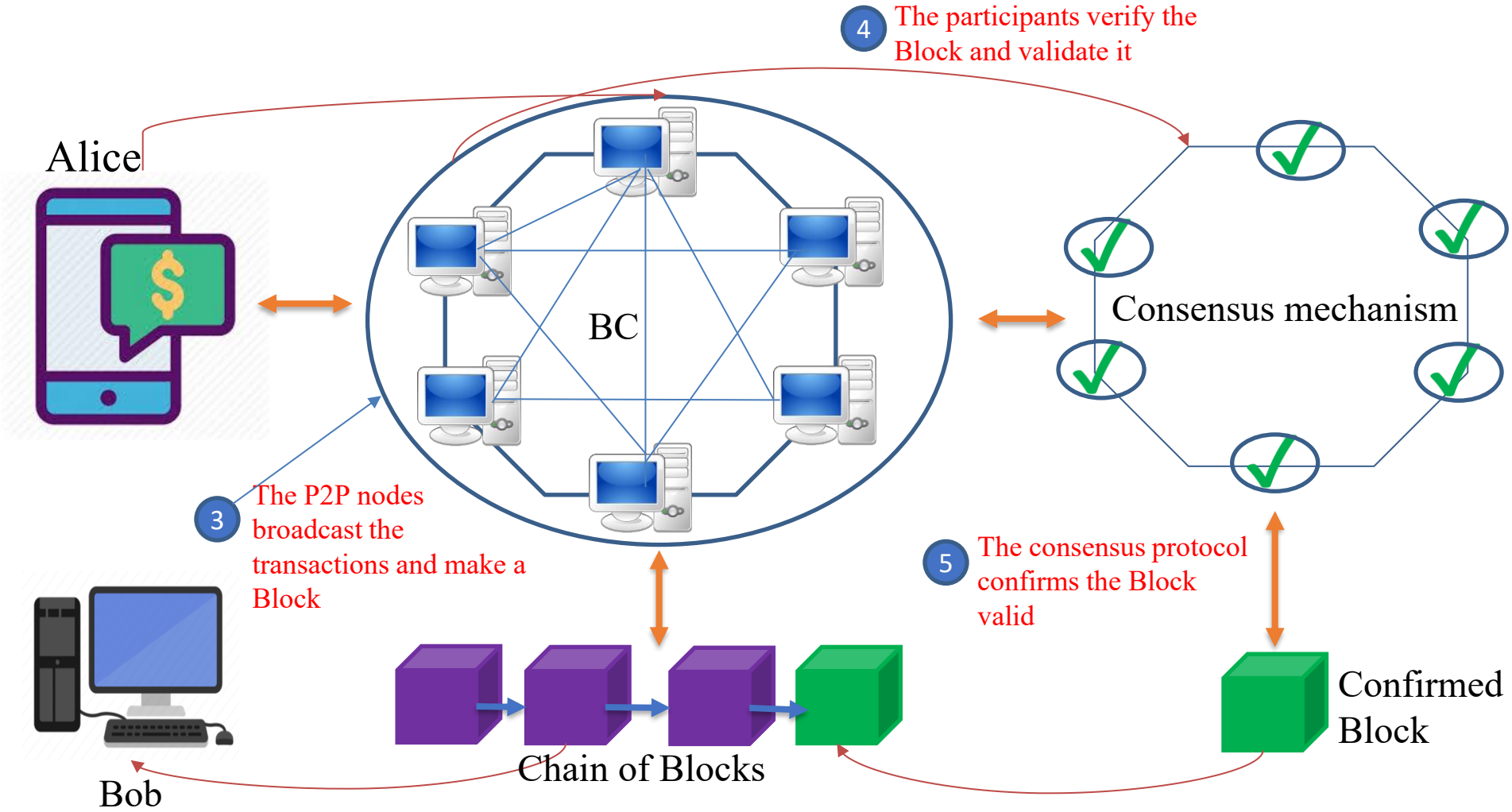
How Does a Blockchain Work?

- The working of blockchain is straightforward.
- **User initiates transaction** - Transaction is created and digitally signed
- **Transaction broadcast** - The individual transaction is sent to network nodes
- **Mempool inclusion** - Transaction enters the pending pool at each node
- **Block creation** - Miners select multiple transactions from the mempool to include in a candidate block
- **Mining process** - Miners compete to solve the cryptographic puzzle for their candidate block
- **Block validation** - Other nodes verify the winning block and its transactions
- **Chain addition** - The validated block is added to the blockchain
- **Transaction confirmation** - The transaction is now confirmed and immutable

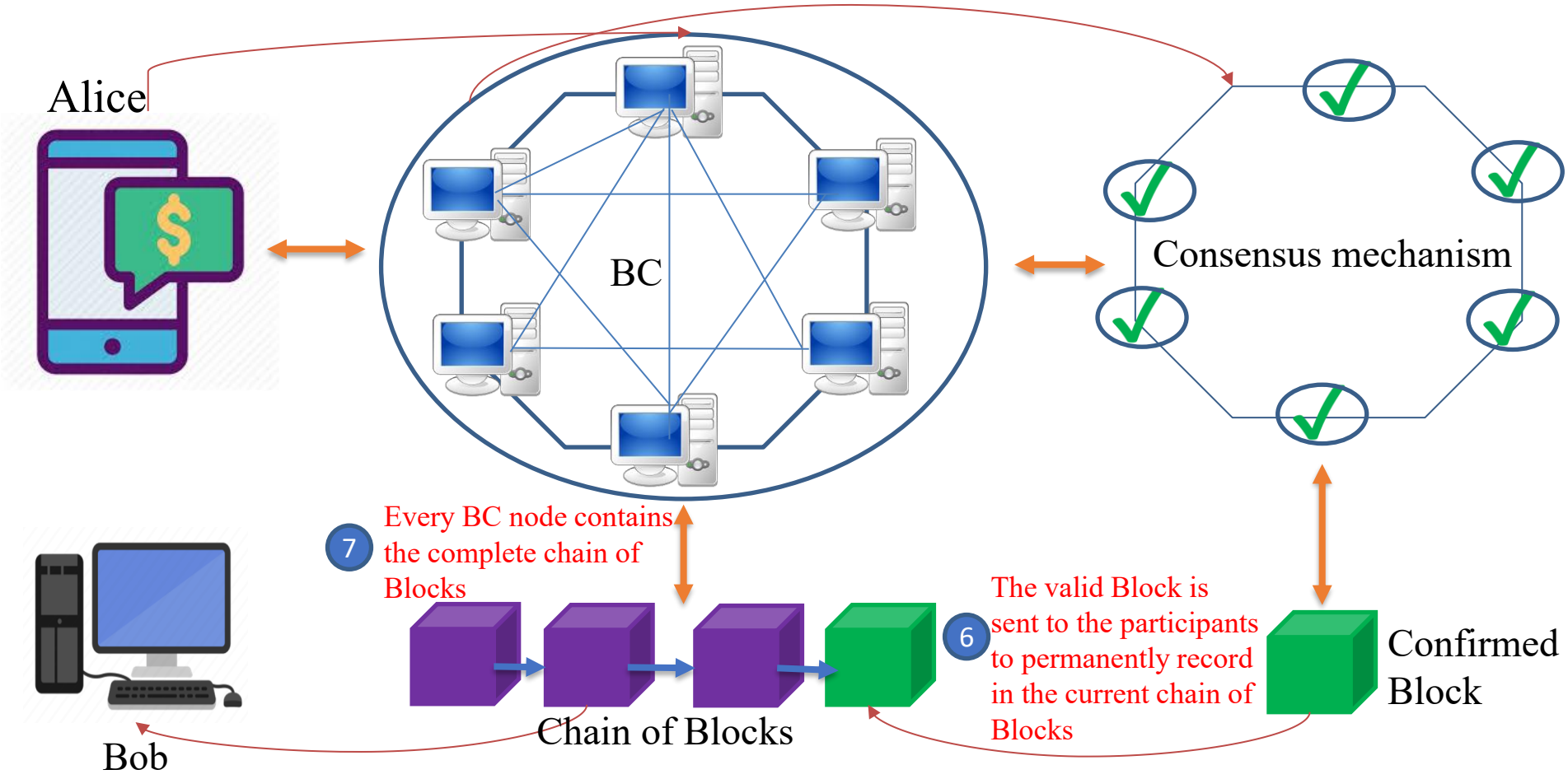
Blockchain: Basic operation



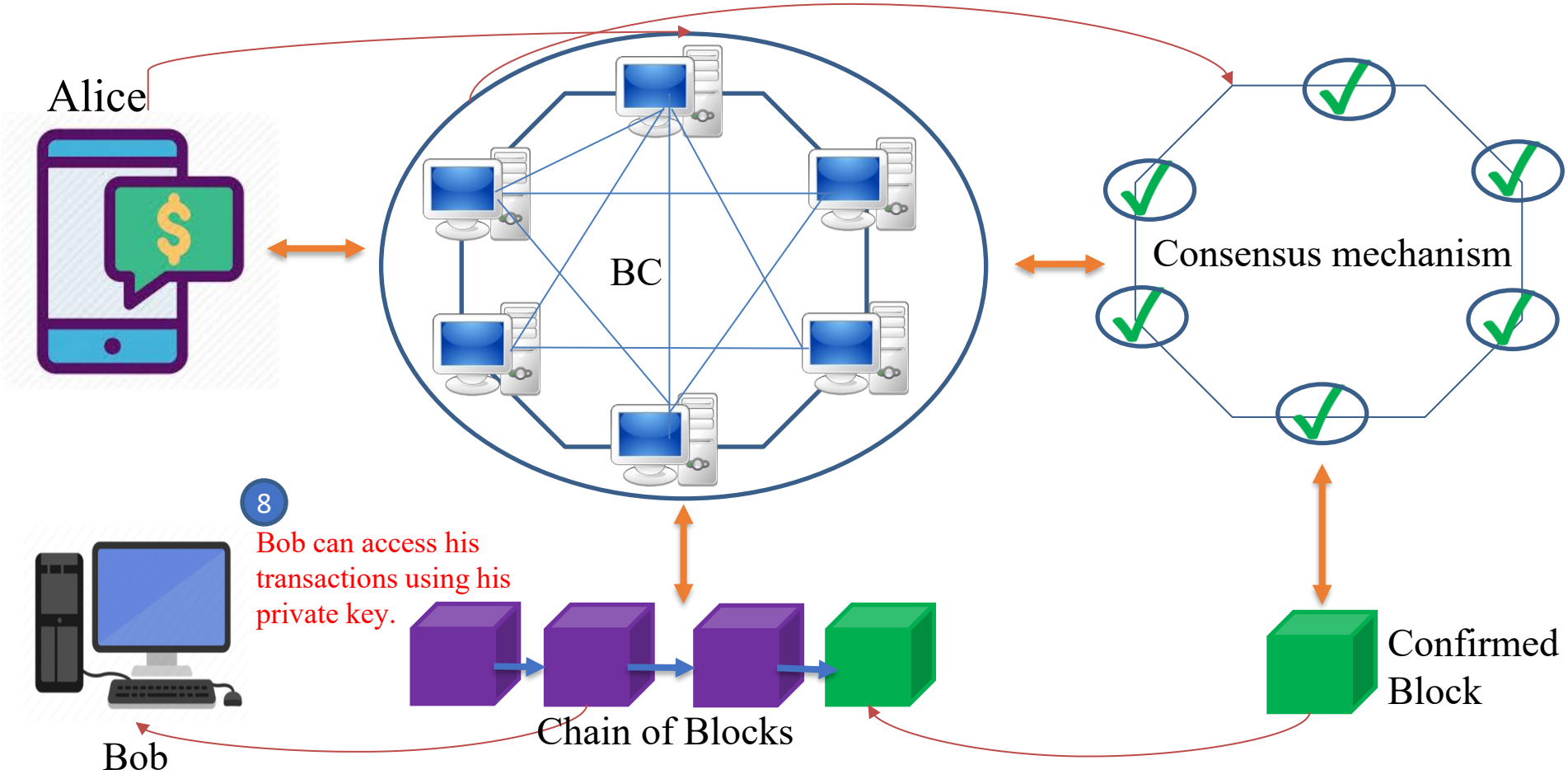
BC: Basic operation



Blockchain: Basic operation



Blockchain: Basic operation

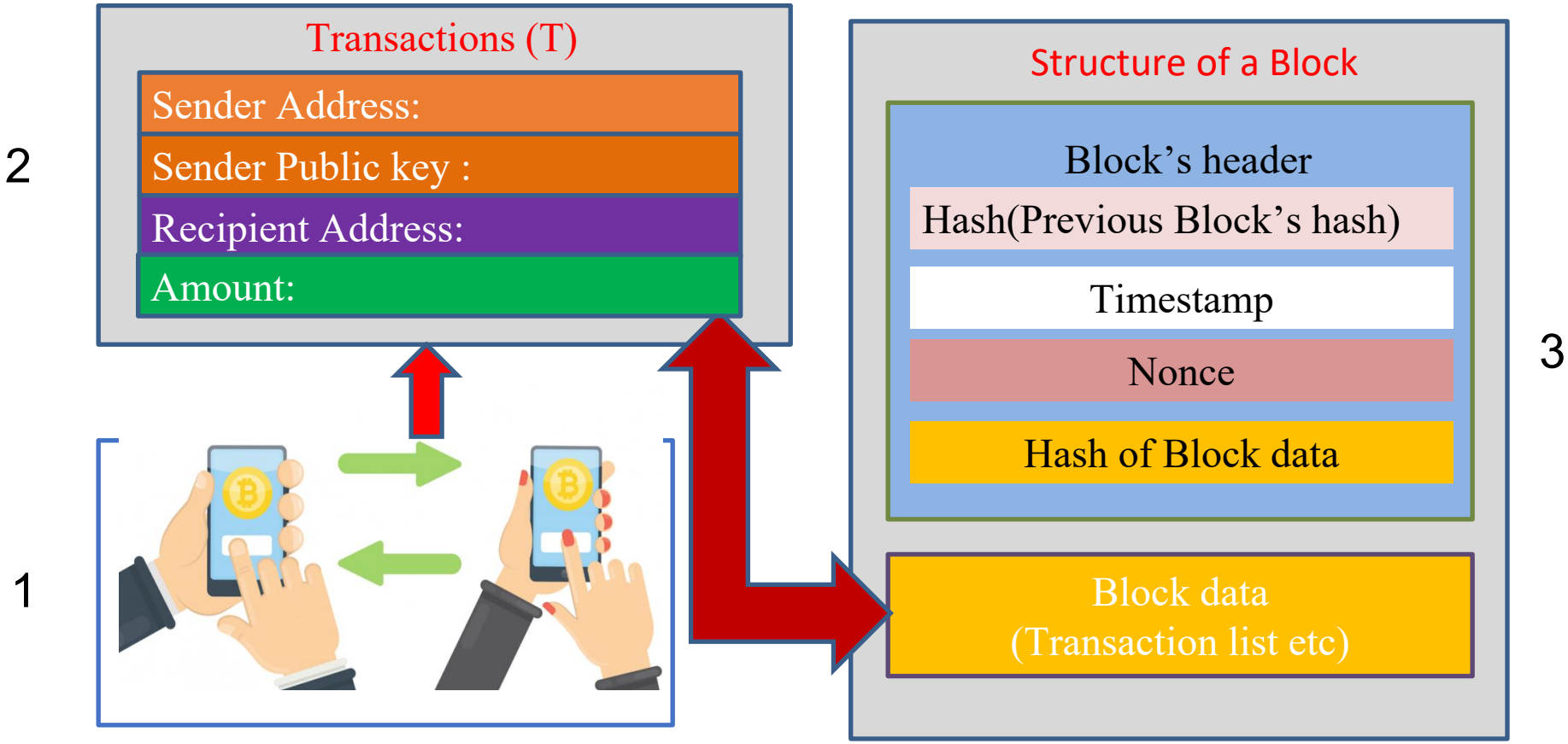


How Does a Blockchain Work?

The functioning of the blockchain is dependent on three components:

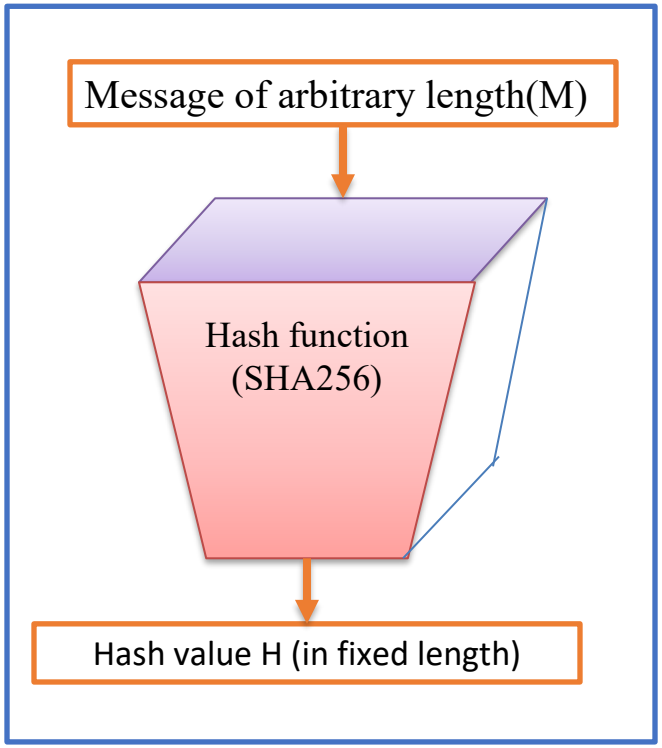
- **Step 1 - Data**:
  - The type of data stored in a block differs based on the blockchain.
  - If the data is about Bitcoin, the blockchain maintains information about a transaction such as a sender, receiver, and transaction amount.
- **Step 2 - Hash**:
  - A hash, similar to a fingerprint, is also included in the block.
  - When a block is created, its hash is computed.
  - If something changes within the block, the hash will change as well.
  - That is why hashing aids in detecting changes in a blockchain.
  - If a block's fingerprint changes, it is no longer the same block.
- **Step 3 - Previous data hash**:
  - In a blockchain, each block is linked to the previous hash's data.
  - if any of the blocks' data is tampered with, the hash changes instantly, and the subsequent block recognizes it and makes the changes immediately.

Data and Block



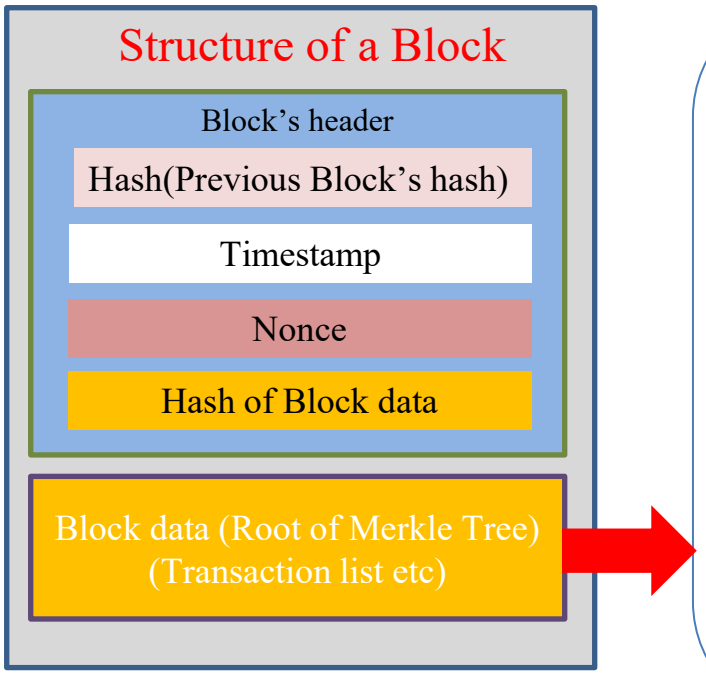
Hash

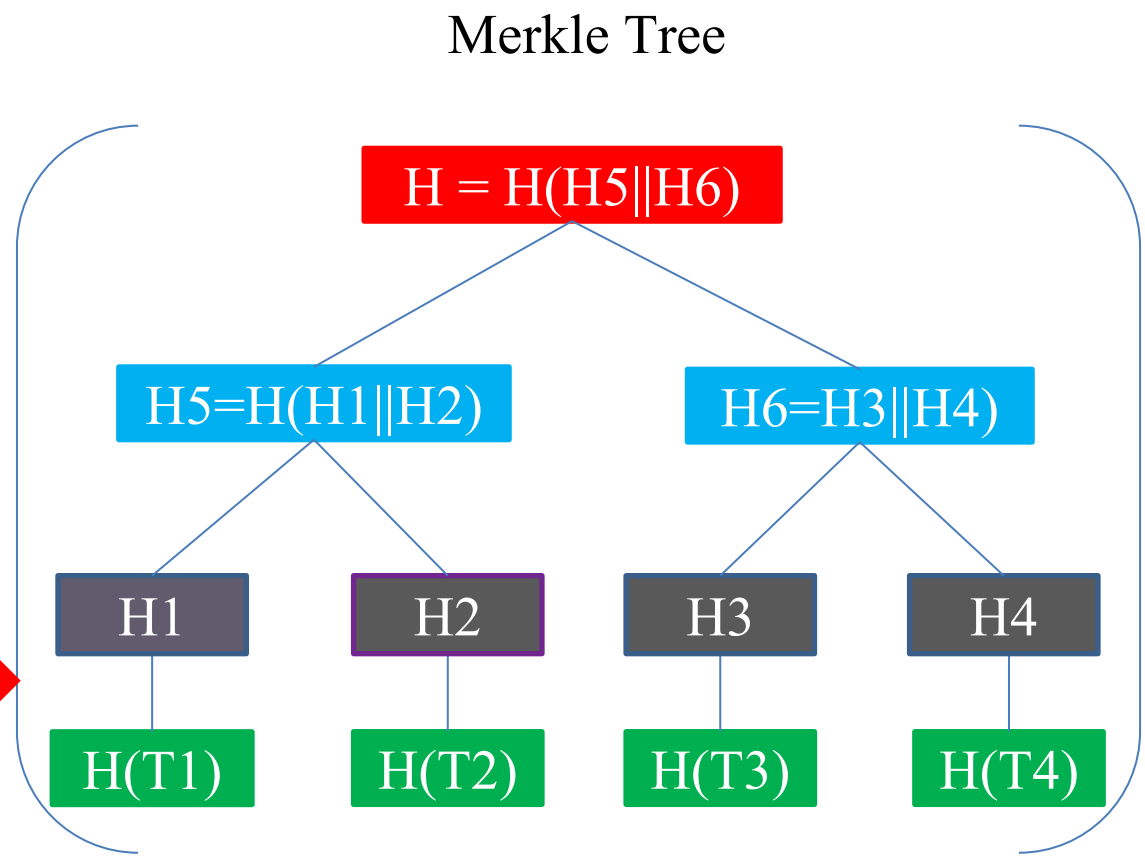
- **Hashes**
  - A mathematical function that takes input data of any length and produces output of fixed length that represents the fingerprint or hash of the data.
- **Two relevant properties of a good hash function are:**
  - hard to calculate back the original data from the hash
  - If the input data changes, the hash changes in an unpredictable way



Simulation

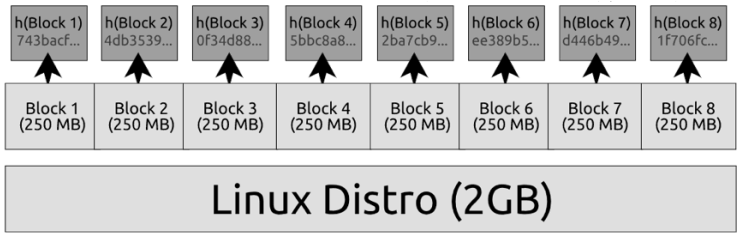
Data Layer: Merkle Tree





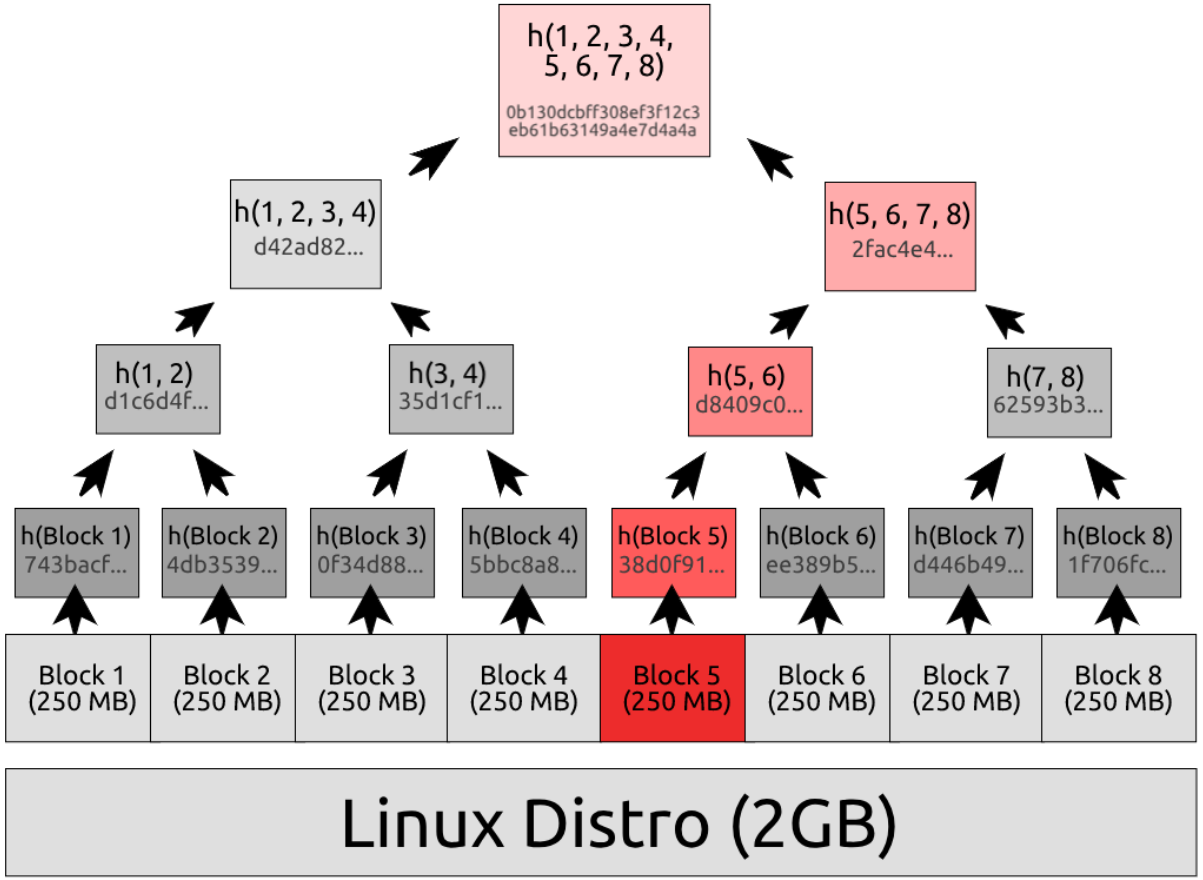
A tree-like data structure used for efficiently verifying the integrity of large sets of data

Merkle Tree



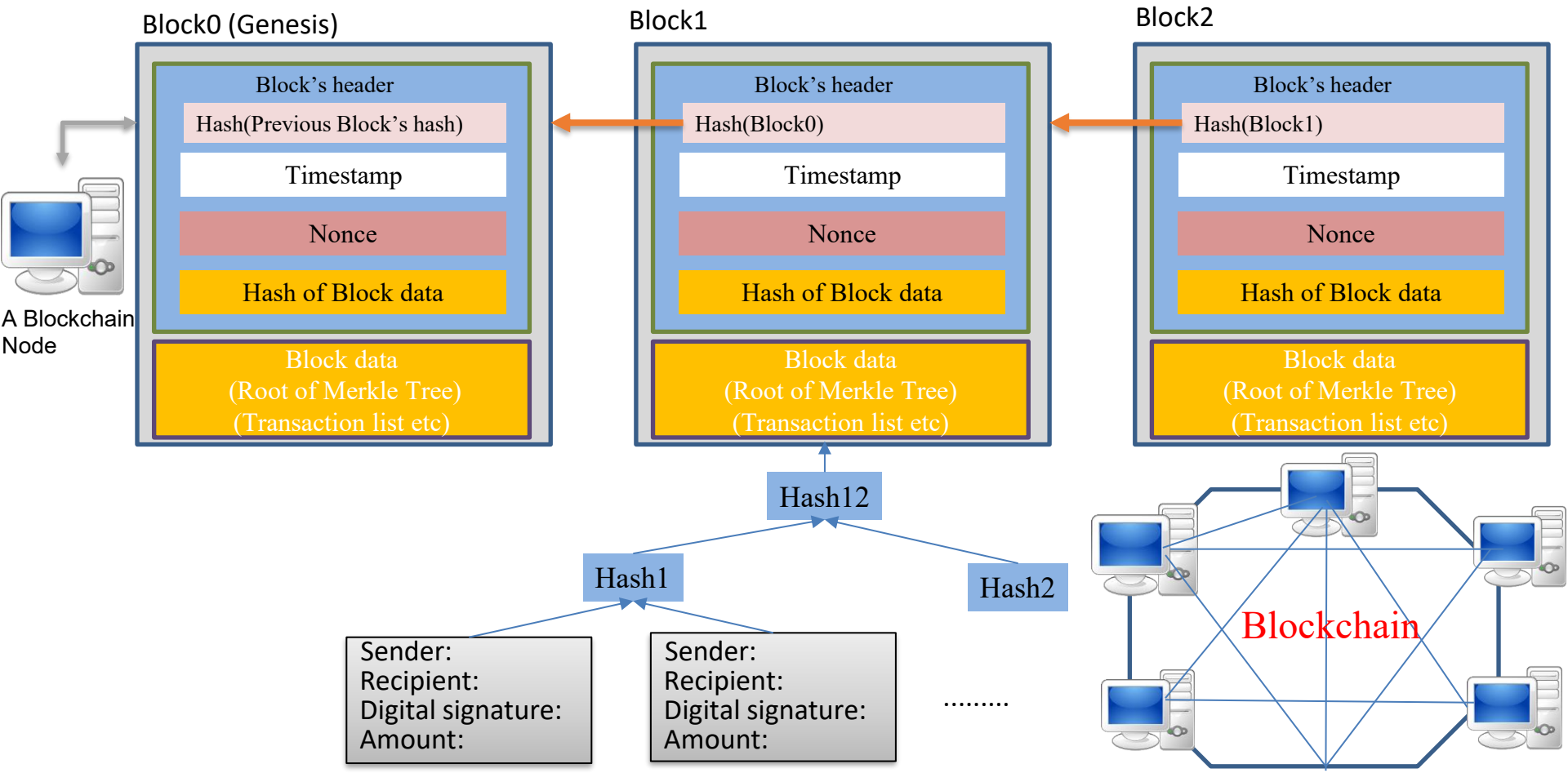
Source

Merkle Tree

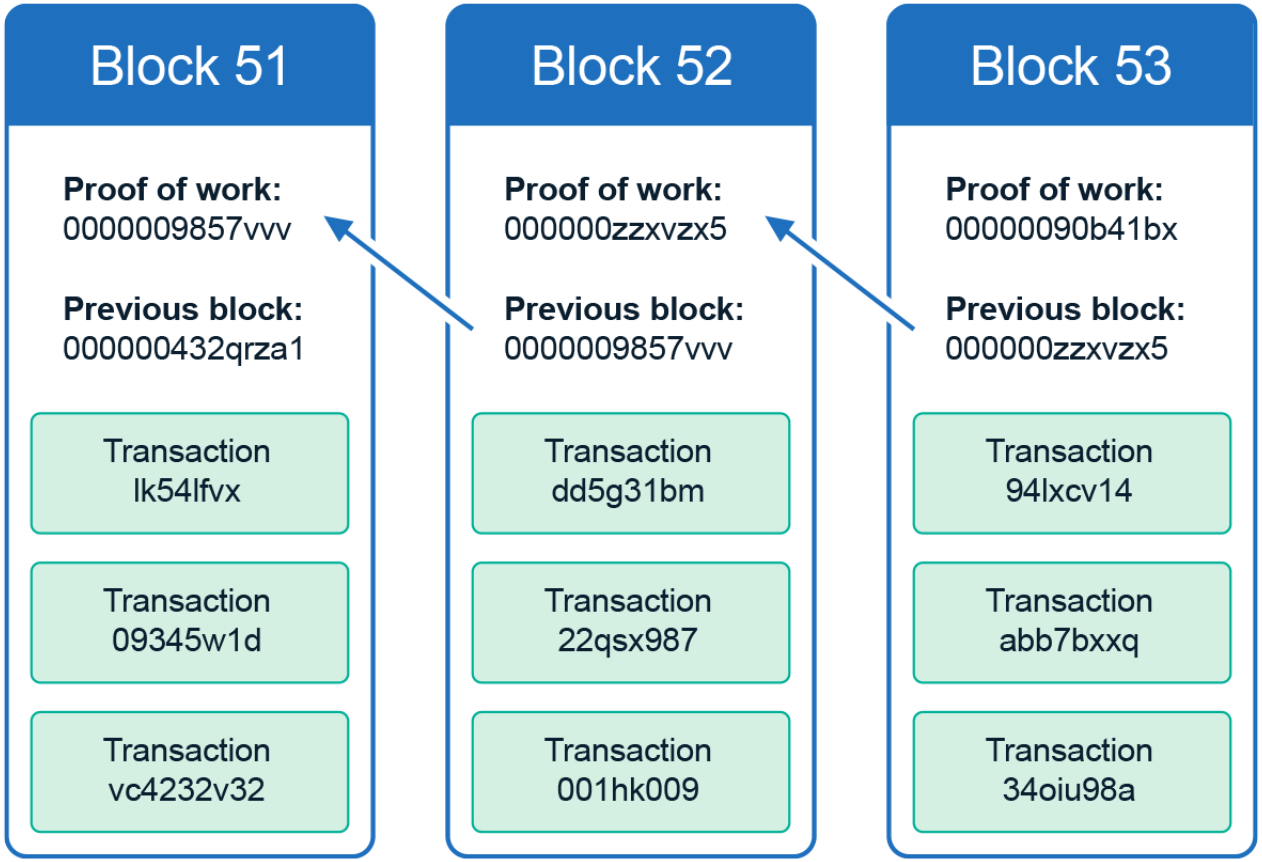


Source

Chain of Blocks: Linked List of blocks

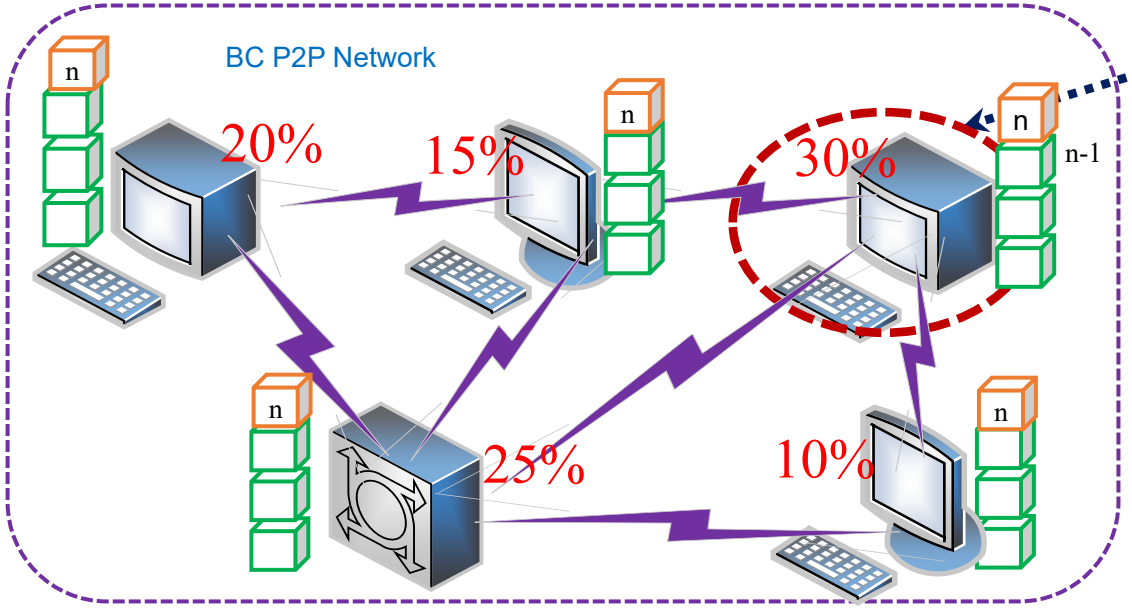


Chain of Blocks



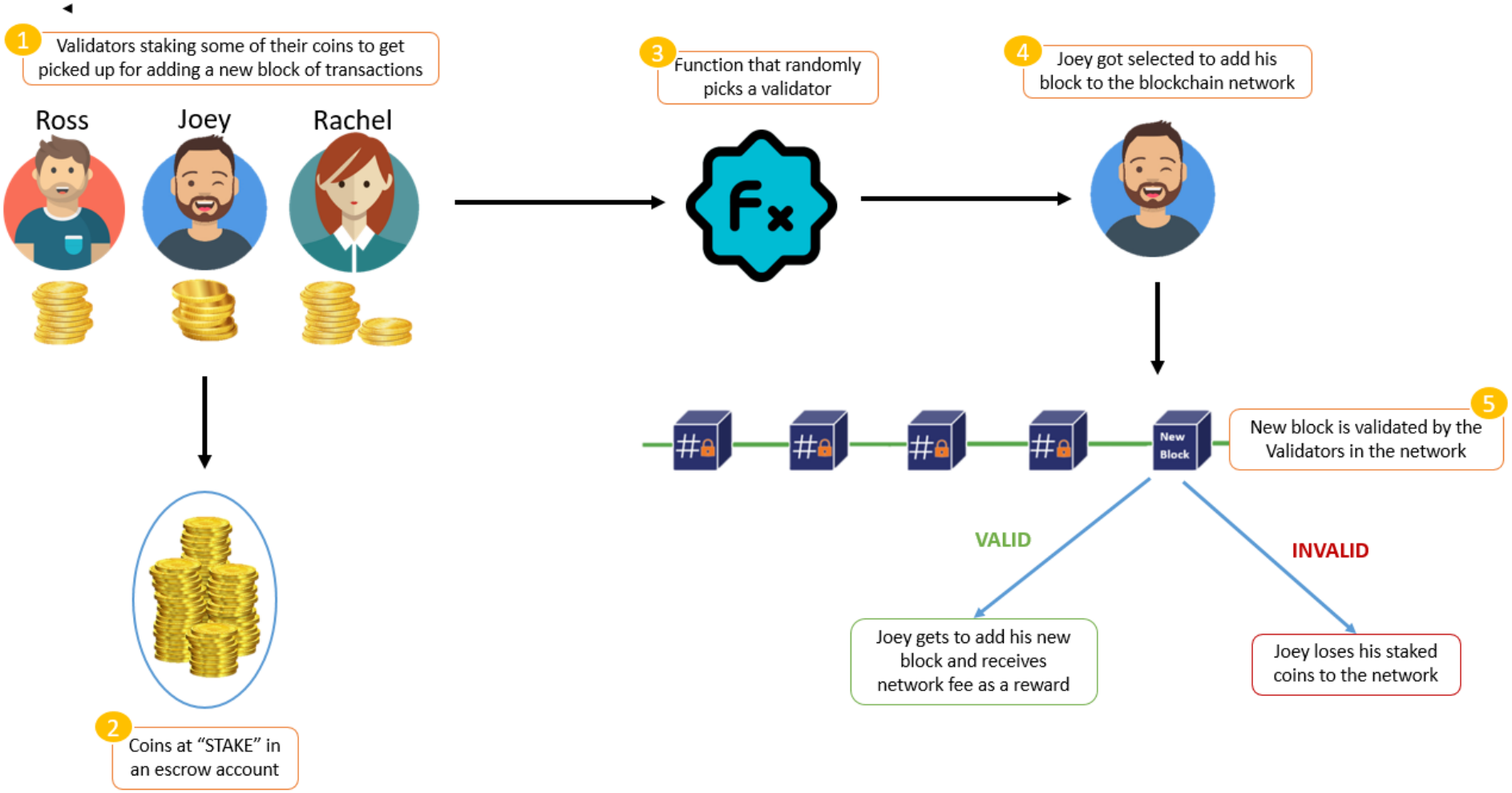
Consensus Protocol: Proof of Stake (PoS)

- The Miner with the higher share make the next Block and receives a reward for this.
- Disadvantages:
  - few rich nodes always dominate,
  - less decentralized,
  - vulnerable to 51% attack.

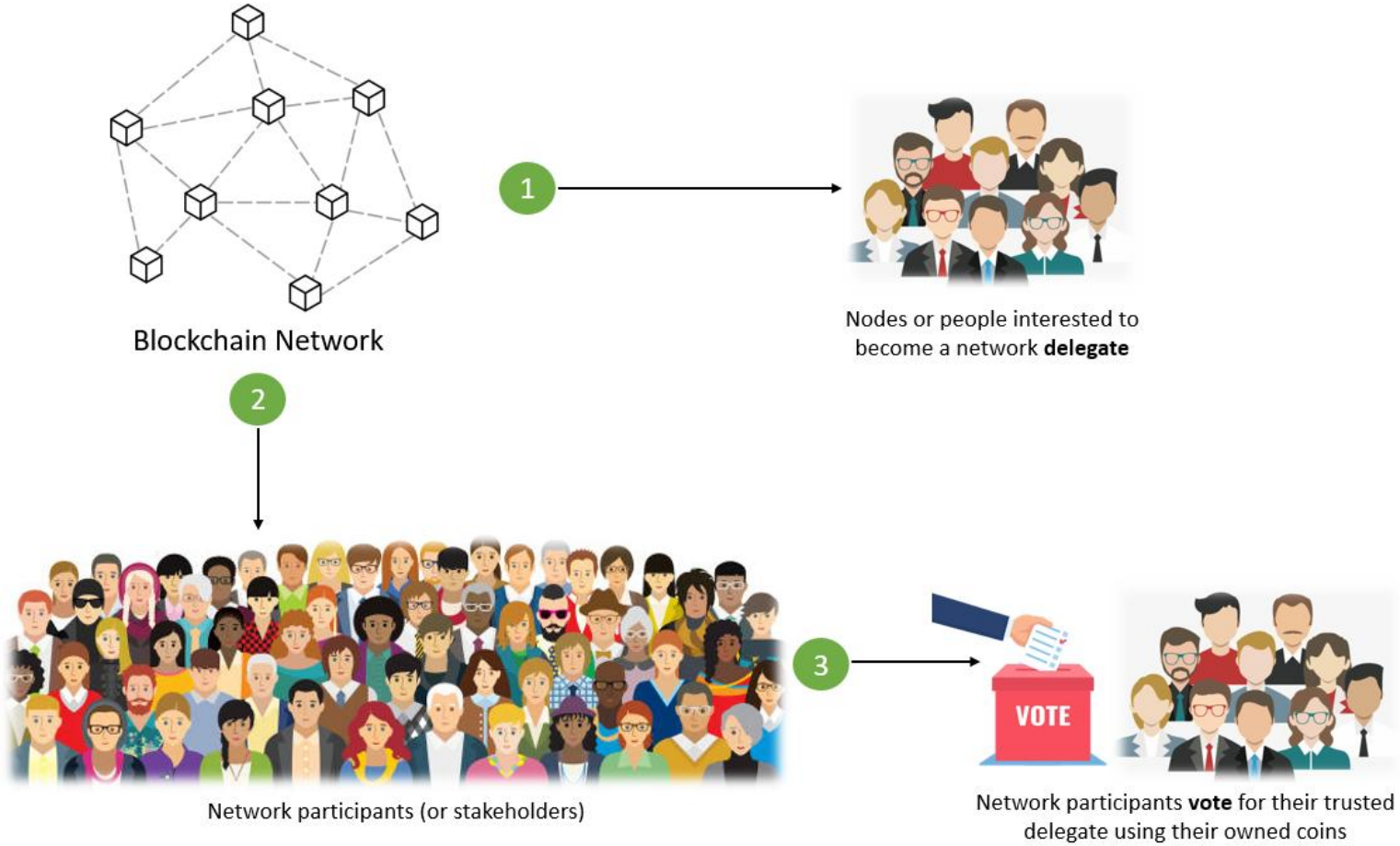


This Miner organize transactions and make block to broadcast the Block throughout the network. Other nodes agree on it.

Proof of Stake

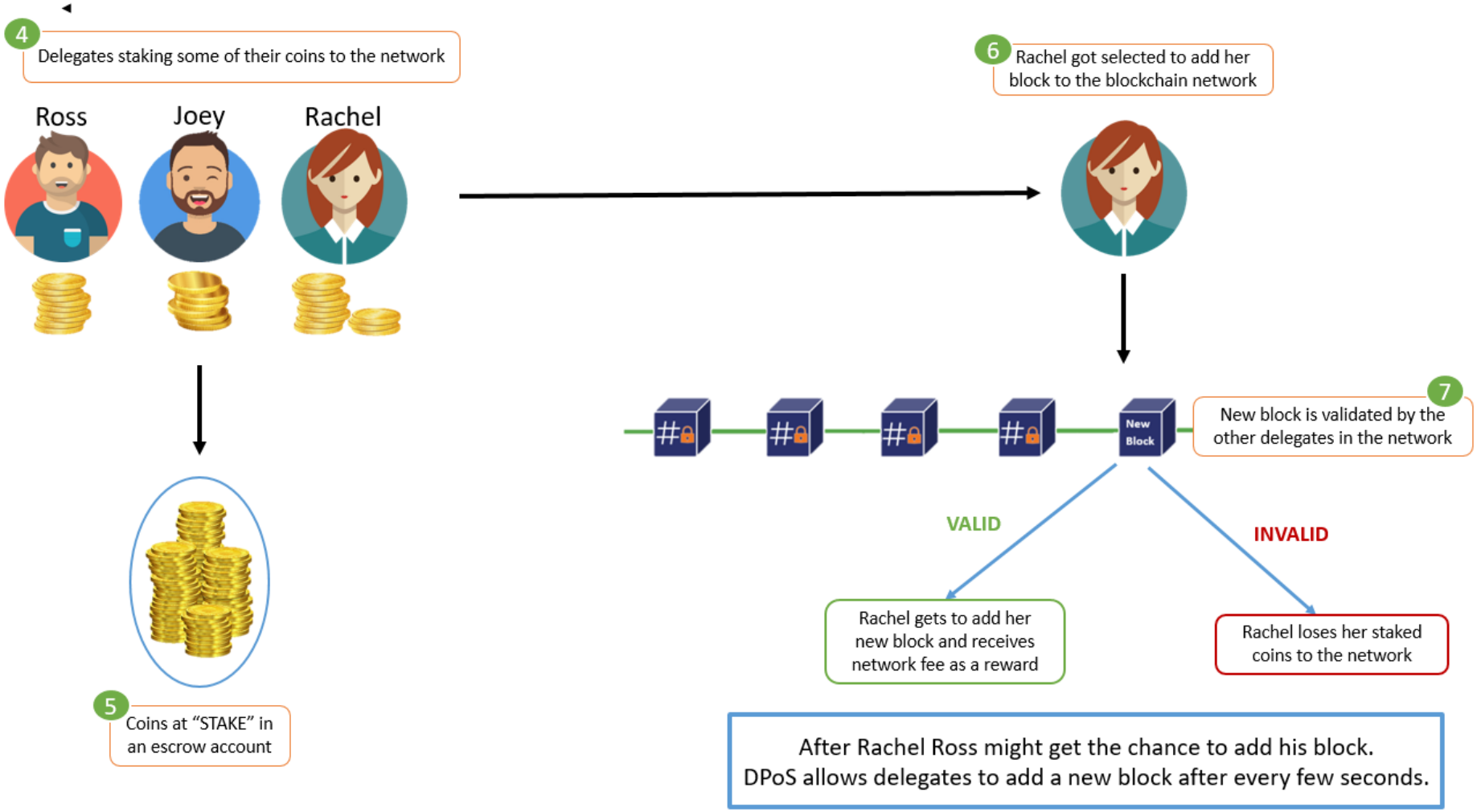


Delegated Proof of Stake



- Users can either directly vote or give their voting power to another entity to vote on their behalf
- Selected witness are responsible for creating blocks by verifying transactions.
- Witness gets reward for verifying transactions

Delegated Proof of Stake



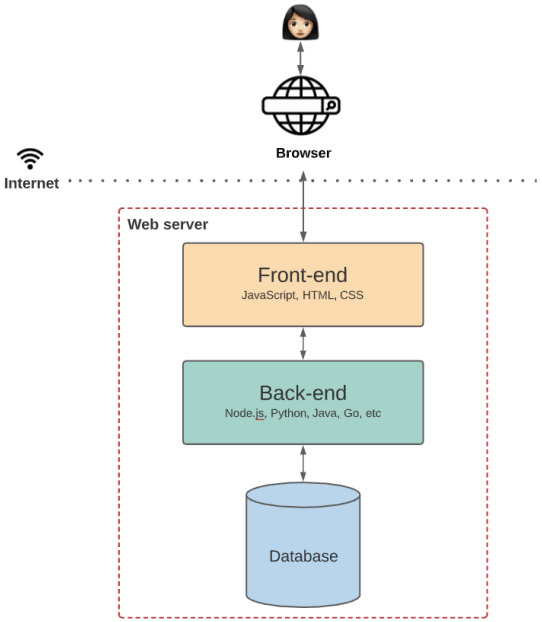
Source

How Does a Blockchain Work?

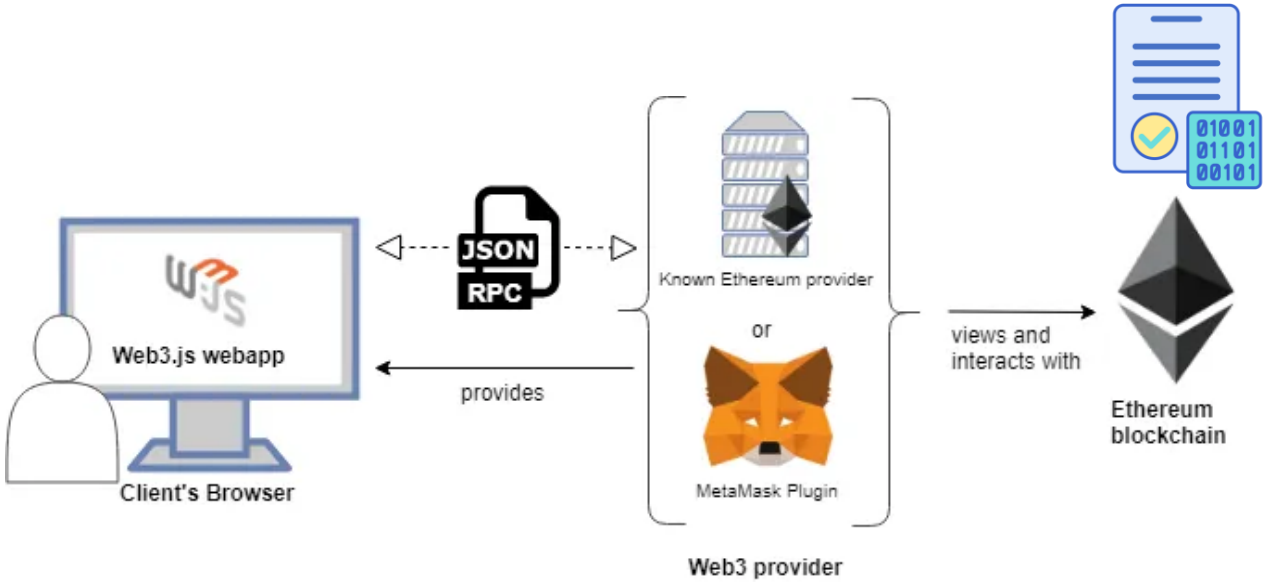
Blockchain Application

- The Blockchain application layer includes **smart contracts (chaincode)**, and **dApps**.
- **A smart contract** refers to a set of business logic presented in various functions that are **executed when** a transaction against those functions is issued.
- **dApps:** a dApp refers to **a distributed web application** that runs on top of a distributed Blockchain technologies such as Ethereum, Bitcoin, and Hyperledger Fabric.
  - dApp <u>can interact</u> with Blockchain using smart contract or chaincode.
  - Unlike a conventional app, dApp is <u>no longer controlled by a single entity</u> or an organization once it is deployed on the Blockchain network.

dApp


Browser
Internet
Web server
Front-end
JavaScript, HTML, CSS
Back-end
Node.js, Python, Java, Go, etc
Database

Web2


Web3.js webapp
Client's Browser
JSON
RPC
provides
Known Ethereum provider
or
MetaMask Plugin
Web3 provider
views and
interacts with
01001
01101
00101
Ethereum
blockchain

Smart contracts

- **Smart contracts** are digital contracts stored on a blockchain that are automatically executed when predetermined terms and conditions are met.
- They typically are used to automate the execution of an agreement so that all participants can be immediately certain of the outcome, without any intermediary's involvement or time loss.
- They can also automate a workflow, triggering the next action when conditions are met.

Smart contracts

- Within a smart contract, there can be as many stipulations as needed to satisfy the participants that the task will be completed satisfactorily.
- To establish the terms, participants must **determine** how transactions and their data are represented on the blockchain, **agree** on the *"if/when...then..."* rules that govern those transactions, **explore** all possible exceptions, and **define** a framework for resolving disputes.
- Then, the smart contract can be programmed by a developer
  - Templates, web interfaces, and other online tools can be used to simplify structuring smart contracts.

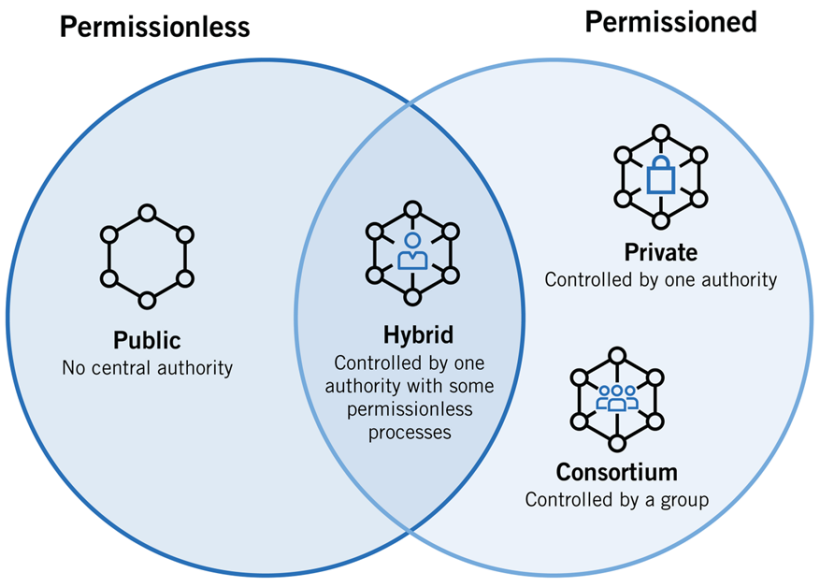
Benefits of smart contracts

- Speed, efficiency and accuracy
- Trust and transparency
- Security
- Savings

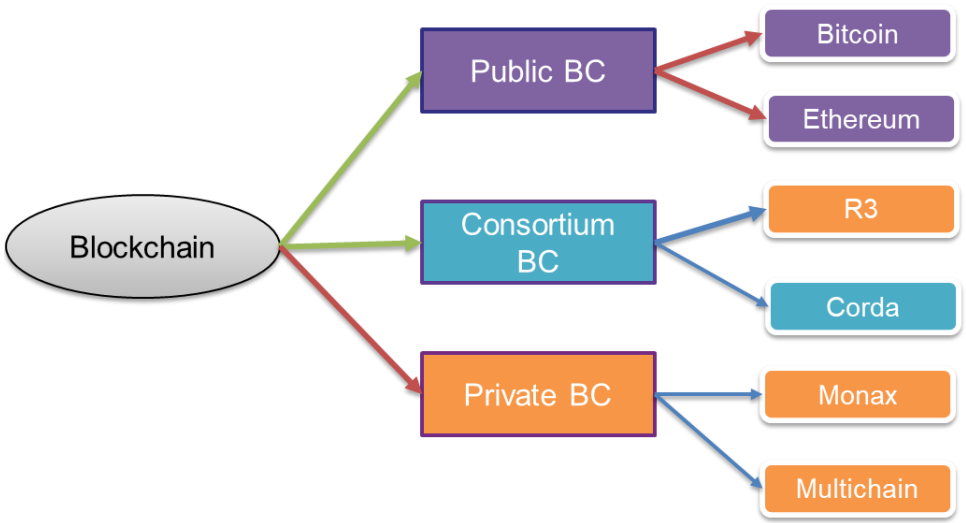
Types of Blockchains

- **Public blockchain**: is one where anyone can participate without restrictions.
  - Most types of cryptocurrencies run on a public blockchain that is governed by rules or consensus algorithms.
- **Permissioned or private blockchain:**
  - Allows organizations to set controls on who can access blockchain data.
  - Only users who are granted permissions can access specific sets of data.
  - Oracle Blockchain Platform is a permissioned blockchain.
- **Federated or consortium blockchain:**
  - closely controlled by a preselected set of nodes or by a preselected number of stakeholders.

Types of Blockchains

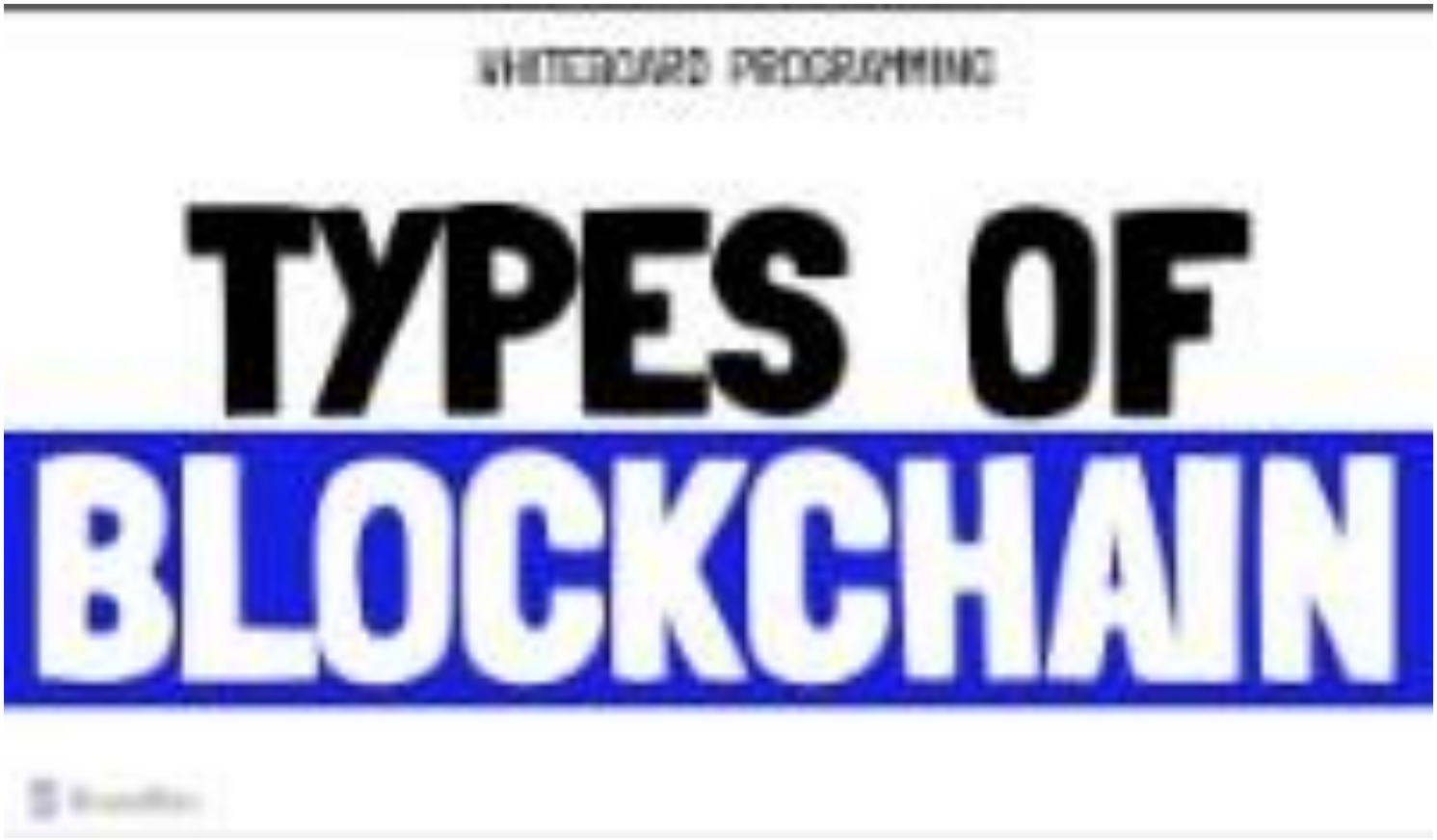


Source



Source

Types of Blockchains



Blockchain digital transformation

- **Blockchain** increases trust, security, transparency, and the traceability of data shared across a business network — and delivers cost savings with new efficiencies.
- **Its impacts** will be **game-changing**, the technology can revolutionize lending, security, consumerism, business models and digital property.
- **Blockchain** holds the promise of transactional transparency – the ability to create secure, real-time communication networks, enabling digital currencies.
- **Blockchain** will begin Web 3.0 which is the 3rd generation of the internet.
  - The internet will take advantage of the blockchain network and bring a truly decentralized network that is way more advanced than what we have right now.

Interesting facts about blockchain

- As of 2021, the market size of the Blockchain industry is at **$4.9 billion**. It's projected to grow to **$67.4** billion by 2026.
- Only **0.5% of the population** are using Blockchain technology now
- **90% of U.S. and European banks and financial institutions** have started exploring blockchain technology
- Many blockchain centric companies are becoming hugely successful
- More **than 20 countries** have actively researched Blockchain

Become a blockchain developer

Digital Currencies

- Motivation for digital currency
- Popular digital currencies in 2024
- Use of digital currencies
- Advantages from digital currencies
- Types of digital currencies
- Governance of digital currencies
- Will we get a boom or doom of digital currencies

Motivation of Digital Currency

Medium of Exchange

- Medium of exchange- bartering is a direct trade of goods and services.
- In modern time, Money allows people to trade goods and services indirectly.

Motivation for digital currency

- **Physical cash**
  - Non-traceable (well, mostly!)
  - Secure (mostly)
  - Low inflation
  - Can’t be used online directly
- **Electronic credit or debit transactions**
  - Bank sees all transactions
  - Merchants can track/profile customers.
- **Cons of Bank**
  - Not peer to peer (Bank must be online during every transaction)
  - Bank can fail
  - Bank can delay or censor transactions
  - Privacy

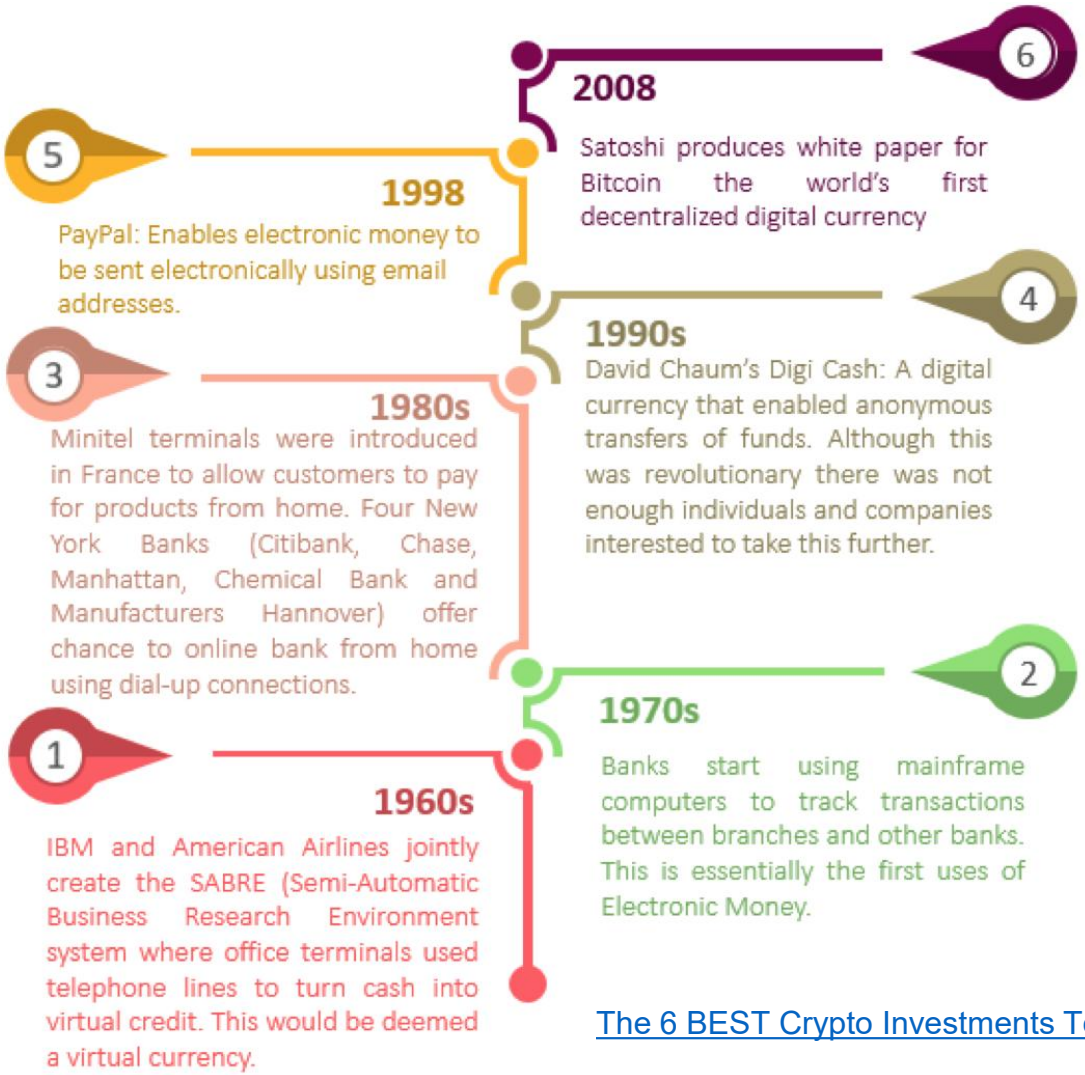
Cryptocurrency

- **Cryptocurrency**
  - A currency that's stored completely digitally.
- A central authority is not involved.
- Make secure with cryptography, distributed consensus, and economic incentive alignment.
- Key characteristics of cryptocurrency
  - *Durability*: does not lose value and not destroyed or made irredeemable easily.
  - *Portability*: easy to transport from place to place.
  - *Divisibility*: easily exchanged in different denominations.
  - *Uniformity*: All units of the currency are identical in value.

Bitcoin: a new internet currency

- Introduced in this world by Satoshi Nakamoto in 2008
  - Satoshi Nakamoto holds is worth about $19.4 billion
- Bitcoin: A Peer-to-Peer Electronic Cash System
- Underlying architecture is the Blockchain
- Decentralized, pseudo-anonymous transfers of value between individuals and businesses.

Digital Currency before Bitcoin



Top 4 popular digital currencies in 2022

1. Bitcoin (BTC)
   - Price: $59,049.10
   - Market cap: $ 1,129.58 billion
2. Ethereum (ETH)
   - Price: $3,066.19
   - Market cap: $368.57billion
3. Tether (USDT)
   - Price: $1.00
   - Market cap: $66 billion
4. USD Coin (USDC)
   - Price: $1.00
   - Market cap: $56 billion

The 6 BEST Crypto Investments To 10x In 2024 (as Bitcoin is Crashing)

https://www.forbes.com/advisor/au/investing/cryptocurrency/top-10-cryptocurrencies/

Bitcoin: Challenges

- **Validation**
  - Is the coin legit? (proof-of-work)

    Use of Cryptographic Hashes
  - How do you prevent a coin from double-spending?

    Broadcast to all nodes
- **Creation of a virtual coin/note**
  - How is it created in the first place?

    Provide incentives for miners, earn bitcoins after work!
  - How do you prevent inflation? (What prevents anyone from creating lots of coins?)

    Limit the creation rate of the BitCoins. Right now, 12.5 coins to miners.

Bitcoin: Challenges

- **Security**
  - **Authentication**
    - Am I paying the right person? Not some other impersonator?
  - **Integrity**
    - Is the coin double-spent?
    - Can an attacker reverse or change transactions?
  - **Availability**
    - Can I make a transaction anytime I want?
  - **Confidentiality**
    - Are my transactions private? Anonymous?

Bitcoin: Challenges

- **Buyer and Seller protection in online transactions**
  - Buyer pays, but the seller doesn't deliver
  - Seller delivers, buyer pays, but the buyer makes a claim.
- **Trust on third-parties**
  - Rely on "proof of work" instead of trust
  - Verifiable by everyone – blockchain is visible to all
  - No central bank or clearing house

Bitcoin Network

- **Each P2P node runs the following algorithm:**
  - New transactions are broadcast to all nodes.
  - Each node (miners) collects new transactions into a block.
  - Each node works on finding a proof-of-work for its block. (Hard to do. Probabilistic. The one to finish early will probably win.)
  - When a node finds a proof-of-work, it broadcasts the block to all nodes.
  - Nodes accept the block only if all transactions in it are valid (digital signature checking) and not already spent (check all the transactions).
  - Nodes express their acceptance by working on creating the next block in the chain, using the hash of the accepted block as the previous hash.

Practical Limitation

- **At least 10 mins to verify a transaction.**
  - Agree to pay,
  - Wait for one block (10 mins) for the transaction to go through,
  - But, for a large transaction ($$$) wait longer, around 60 minutes. Because if you wait longer, it becomes more secure.
  - For large $$$, you wait for six blocks (1 hour).

Privacy Implications

- **No anonymity, only pseudonymity**
- **All transactions remain on the block chain– indefinitely!**
- **Retroactive data mining**:
  - Target used data mining on customer purchases to identify pregnant women and target ads at them (NYT 2012), ended up informing a woman's father that his teenage daughter was pregnant.
  - Imagine what credit card companies could do with the data.

Cryptocurrency Scams

- **Fake websites**
  - Scammers sometimes create fake cryptocurrency trading platforms or fake versions of official crypto wallets to trick unsuspecting victims.
- **Phishing scams**
  - Scammers send an email to lure recipients to a specially created website asking them to enter private key information.
  - Once the hackers have acquired this information, they steal the cryptocurrency in those wallets.
- **Pump and dump schemes**
  - This involves a particular coin or token being hyped by fraudsters through an email blast or social media such as Twitter, Facebook, or Telegram.
- **Cloud mining scams**
  - Cloud mining refers to companies that allow you to rent mining hardware they operate in exchange for a fixed fee and a share of the revenue you will supposedly make.
  - However, many cloud mining companies are scams or, at best, ineffective – in that you end up losing money or earning less than was implied.

How to spot cryptocurrency scams?

- **Promises of guaranteed returns:**
  - No financial investment can guarantee future returns because investments can go down as well as up.
  - Any crypto offering that promises you will definitely make money is a red flag.
- **A poor or non-existent whitepaper:**
  - Every cryptocurrency should have a whitepaper since this is one of the most critical aspects of an initial coin offering.
  - The whitepaper should explain how the cryptocurrency has been designed and how it will work. If the whitepaper doesn't make sense – or worse, doesn't exist – then tread carefully.
- **Excessive marketing:**
  - All businesses promote themselves.
  - But one way that crypto fraudsters attract people is by investing in heavy marketing – online advertising, paid influencers, offline promotion, and so on.
- **Free money:**
  - Whether in cash or cryptocurrency, any investment opportunity promising free money is likely to be fake.

How to protect yourself from cryptocurrency scams?

- Protect your wallet: Keep your wallet keys private
- Keep an eye on your wallet app
- Only invest in things you understand
- Take your time
- Be wary of social media adverts
- Ignore cold calls
- Only download apps from official platforms
- Do your research

Use of digital currencies

- Digital money is money in purely digital form. It is not a physically tangible asset like cash or other commodities like gold or oil.
- Digital money can streamline the current financial infrastructure, making it cheaper and faster to conduct monetary transactions. It can also ease monetary policy implementation by central banks.
- Examples of types of digital money are cryptocurrencies, central bank digital currencies, and stable coins.
- Digital money is susceptible to hacks and can compromise user privacy.

Advantages from digital currencies

- Reduce the time lag and operating costs for such transactions by using distributed ledger technology (DLT)
  - In a DLT system, nodes or shared ledgers connect to form a common network to process transactions.
  - This network can also extend to other jurisdictions and minimize the processing time for transactions.
  - It provides transparency to authorities and stakeholders, improving the resiliency of a financial network by eliminating the need for a centralized database of records.
- Solve the double-spending problem by using an algorithmic consensus system.
  - The problem, stated quite simply, relates to ensuring that a "note" of digital money is not spent twice by the same person.

Types of digital currencies

- **Central Bank Digital Currencies**
  - replicate the role of a central authority in ensuring solvency and integrity of transactions, albeit in a digital context.
- **Cryptocurrencies**
  - Eliminate the function of central authorities
  - Blind signatures hide the identity of transacting parties
  - Zero-knowledge proofs encrypt transaction details
- **Stablecoins**
  - a variation of cryptocurrencies and were developed to counter the price volatility of regular cryptocurrencies
  - A form of private money whose price is tied to that of a fiat currency or a basket of goods to ensure that they remain stable.

3 types of digital currencies

Governance of digital currencies

- **Regulatory and policy gaps and inconsistencies of digital currencies:**
  - A risk-based regulatory approach can provide more flexibility to accommodate future innovation.
  - While complete future-proofing is impossible in such an ever-evolving landscape for digital currency, premature regulations could stifle productive innovation and limit societal benefit.
- **Privacy and confidentiality options for central bank digital currencies:**
  - "Privacy by design" can be integrated with "security by design" to enable higher CBDC adoption and responsible deployment.
- **Digital currency consumer protection risk mapping:**
  - Consumer education will be a critical component in ensuring that consumers can make informed decisions that match their needs without exposing them to undue risk.

Governance of digital currencies in EU

- **Digital Currencies Governance Group** (**DCGG**) is an international trade association and consultancy, created to strengthen the voice of the digital assets industry.
- **DCGG's** mission is to **ensure** that any and all digital asset-related regulations or policies support rather than stifle technology, innovation and growth.
- **DCGG** supports the EU's Digital Finance Strategy and the accompanying legislative proposals that seek to **ensure a robust EU financial sector**,
  - which services to modernise the European economy and give consumers access to innovative financial products, while ensuring consumer protection and financial stability.

Governance of digital currency in US

- While the US Federal Reserve has made no decisions on whether to pursue or implement A Central Bank Digital Currency, or **CBDC**, we have been exploring the potential benefits and risks of **CBDCs** from a variety of angles, including through technological research and experimentation.
- **CBDC** is generally defined as a digital liability of a central bank that is widely available to the general public.
- Today in the United States, Federal Reserve notes are the only type of central bank money available to the general public.
- Like existing forms of money, a CBDC would enable the general public to make digital payments. As a liability of the Federal Reserve, however, a CBDC would be the safest digital asset available to the general public, with no associated credit or liquidity risk.

Future of digital currency

- Cryptocurrency regulation
- Crypto ETF approval
- Broader institutional cryptocurrency adoption
- Bitcoin’s future outlook
- Future of cryptocurrency

Reference

- Youtube
- https://www.oracle.com/middleeast/blockchain/what-is-blockchain/
- https://blogs.idc.com/2020/12/14/how-blockchain-plays-into-digital-transformation/
- https://www.analyticssteps.com/blogs/what-blockchain-and-how-does-it-work
- https://www.kdnuggets.com/2017/11/blockchain-key-terms-explained.html
- https://101blockchains.com/blockchain-fundamentals-presentation/
- https://medium.datadriveninvestor.com/7-crucial-facts-on-blockchain-you-must-know-now-a29341ff1f7d